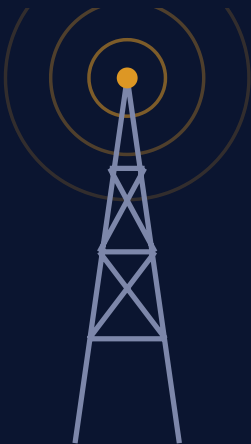


JioMausam

India’s biggest weather network is already built.
We just have to listen to it.

This brief explains the whole idea in plain language: the problem, the science, the product, the working prototype, the business case and the risks. Read it once end to end. Then use the last pages as your cheat sheet for the jury’s questions.

Working prototype: claude.ai/artifact/Wa5Cqt1S6Y61CATcN9SZCK
Private link. Share it from the page’s Share menu before sending it to anyone.



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The idea in one page

JioMausam turns Jio's mobile network into a street-level weather sensor. Mausam Kavach uses that data to pay workers automatically, within hours, when extreme rain or heat stops their work.

PROBLEM

Weather hits one street, not one state. India has about 40 Doppler radars for 3.3 million km², so most extreme rain and heat is never measured where it lands. The people who lose a day's income (riders, vendors, labourers, farmers) have no cover. When insurance exists, claims take months.

INSIGHT

Rain weakens the radio signal between two towers, and the size of the drop tells you how hard it is raining. Networks already log this every few minutes to monitor themselves. Phone battery temperatures, averaged across a city, track air temperature. **So the network is already a weather sensor.**

SOLUTION

JioMausam: a live rain and heat map at about 1 km and 10-minute resolution, built from network data with no new hardware.

Mausam Kavach: ₹29/month parametric cover added at recharge. When the measured rain or heat crosses a set trigger at your location, ₹300 goes to your JioPay wallet. There is no claim to file.

PROOF (OUR PROTOTYPE)

We ran the method on **500 real microwave links** (11 days, 1-minute data) and scored it against national weather radar. On days it had not been tuned on, network-average rain matched radar at **$r = 0.95$** . It caught **91% of rainy days** with 4.6% false alarms, and our payout rule made the **same decision as radar in 94.6%** of cases.

BUSINESS

Three revenue layers. (1) Kavach premiums: 1% of Jio's 500M users is 5M policies, about ₹170 Cr a year. (2) B2B cover and data for gig platforms, insurers and state agencies. (3) Reliance as customer zero: JioMart demand, New Energy solar forecasts and Jio's own network planning.

WHY ONLY RELIANCE

It needs a national network (sensor), 500M customers who recharge (distribution), JioPay (payout rail), Jio Financial Services' insurance partners (product), and retail and energy businesses that use the data. No startup, insurer or other telecom has all five.

SAY IT IN 10 SECONDS

"Every Jio tower is a rain gauge and every phone is a thermometer. We map India's weather street by street, and pay workers within hours when it stops their work."

How we meet each judging criterion

The competition lists six things the idea should be. Use these answers, in these words, whenever a judge asks “what makes this different?”

CRITERION	OUR ANSWER	EVIDENCE WE CAN SHOW
Different by Design	Everyone else builds new sensors: radars, weather stations, satellites. We build none. The design choice is to reuse a network that already exists, so the sensor is free and already everywhere.	Insight slide; zero-hardware architecture (page 6)
Futuristic	Infrastructure that senses its environment: networks become climate sensors, and insurance pays itself out from live data. This is where 5G/6G “integrated sensing and communication” research is heading.	Prototype map replay; roadmap (page 13)
Solves a real problem	Extreme heat and rain cost informal workers income every year with no protection. Crop and weather claims take months. Cities lack street-level flood and heat data.	Problem page with sources (page 4)
Challenges conventional wisdom	Conventional wisdom says better weather data needs more capex and that micro-insurance can’t pay for itself. We show the data can come from existing assets, and that parametric payouts remove the claims cost that kills small policies.	Results vs radar (page 8)
Practical and scalable	Software on top of existing network logs. It is proven in peer-reviewed research and national-scale pilots abroad. It scales with the network: every new site adds coverage.	Open-source pipeline we ran; pilot plan (page 13)
Strong business case	Recurring premium income with near-zero distribution cost (the recharge), B2B data revenue, and internal savings across Reliance.	Unit economics (page 11)

What JioMausam is not

Not a weather app

The map is a means. The value is the decisions it triggers: payouts, alerts, forecasts for businesses.

Not an insurer

Jio distributes and supplies the trigger data. A licensed insurer underwrites, through Jio Financial Services’ partners.

Not new hardware

No radars, no stations, no towers. If a new device is ever needed, it is a cheap add-on, not the core.

Weather is hyperlocal. Our data isn't.

A cloudburst can flood one ward while the next stays dry. On a heatwave day, temperatures can differ by several degrees between a concrete market and a leafy campus a few kilometres away. Yet India measures weather at a scale of tens of kilometres.

~40

Doppler weather radars for 3.3 million km² (IMD, 2024). Mission Mausam (2024) is spending about ₹2,000 crore to expand observation.

34M

full-time jobs' worth of working hours India could lose to heat stress by 2030 (ILO, 2019).

~90%

of India's workforce is informal, with no paid sick days or income cover when weather stops work.

Who loses, and how

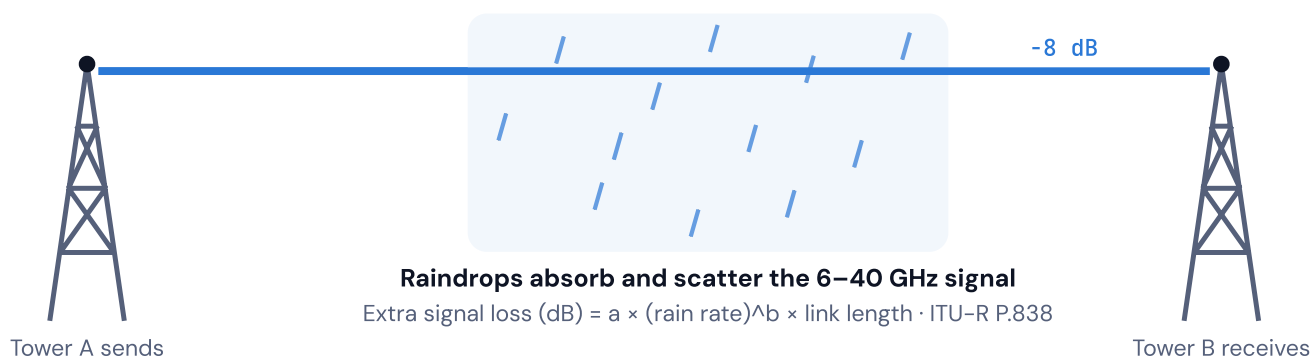
WHO	WHAT EXTREME WEATHER DOES TO THEM	WHAT THEY HAVE TODAY
Delivery riders and gig workers 7.7M in 2020–21, projected 23.5M by 2029–30 (NITI Aayog)	Heavy rain stops orders or makes riding unsafe. Heatwaves cut the hours they can work.	Nothing. Some states have passed gig-worker welfare laws, but there is no weather cover.
Street vendors, construction and farm labour	No customers in a downpour. Work stops above safe heat levels. A lost day is lost income.	Nothing, or informal loans.
Small farmers	Local cloudbursts, dry spells and hail damage crops field by field.	Crop insurance (PMFBY) based on surveys and crop-cutting experiments. Payouts often take months.
Cities and disaster agencies	Urban flooding builds street by street within an hour.	Sparse gauges and radar. Warnings are too coarse to act on locally.

Why existing options fall short

- **Radars** are expensive (each costs crores and needs staff and maintenance). Coverage gaps remain, especially in hills and remote districts.
- **Rain gauges and weather stations** measure a single point and are spread thin. Many report daily, not minute by minute.
- **Satellites** see clouds from above, at roughly 10 km and 30-minute resolution, not what actually reaches the street.
- **Indemnity insurance** needs someone to verify each loss. For a ₹300 lost day, the cost of checking the claim is more than the claim.

The gap in one line: no one has a cheap, dense, real-time record of the weather that hits a specific street. So no one can price that risk or pay for it quickly.

How a mobile network measures rain and heat



Rain: the radio link as a rain gauge

Towers that are not connected by fibre talk to each other over point-to-point microwave links, usually between 6 and 40 GHz. At these frequencies raindrops absorb and scatter energy. The heavier the rain, the bigger the drop in received power.

The physics is standard (ITU-R P.838): extra loss grows as a power of rain rate. Run it backwards and the signal loss gives you rain rate, averaged along the 1–30 km path, minute by minute.

The data already exists. Operators record transmitted and received power on every link to monitor network health. JioMausam reuses those logs.

Heat: phones and sites as thermometers

A phone's battery temperature is mostly body and air temperature. Averaged over many phones in a city, it tracks the daily air temperature closely. This was shown on real data from 8 cities (Overeem et al., 2013).

With user consent, Jio can aggregate anonymised battery readings by area. Many cell sites also log equipment and shelter temperatures, which adds fixed reference points.

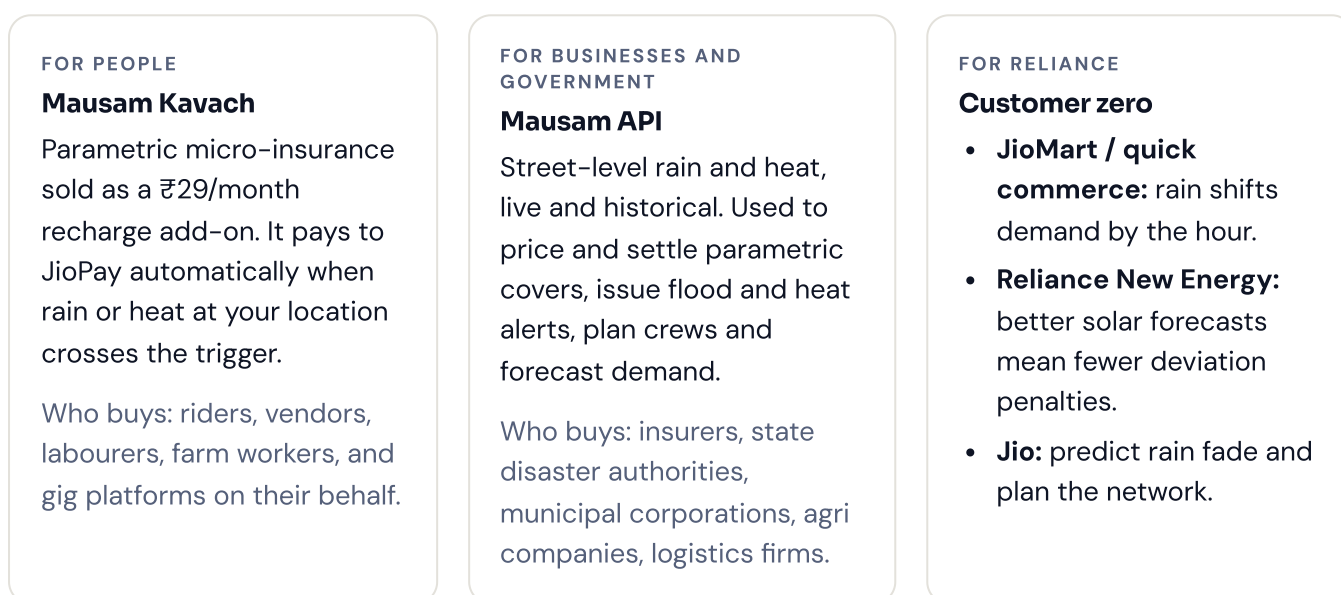
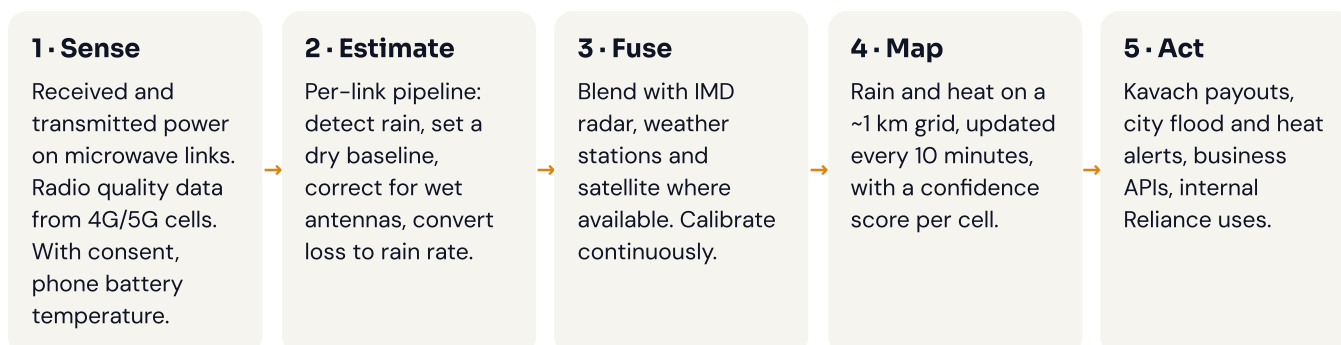
Heat is our second module. Rain is the one our prototype already proves on real data.

This is proven science, not a lab idea

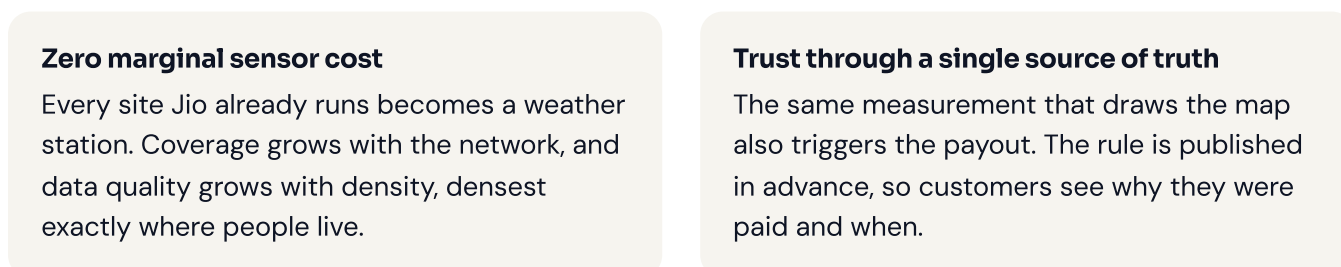
YEAR	WHO	WHAT THEY SHOWED
2006	Messer, Zinevich & Alpert, <i>Science</i>	Mobile-network microwave links measure rainfall accurately (Israel).
2013	Overeem, Leijnse & Uijlenhoet, <i>PNAS</i>	Country-wide rain maps of the Netherlands from one operator's links.
2013	Overeem et al., <i>Geophysical Research Letters</i>	City air temperature from crowd-sourced smartphone battery temperatures.
2016+	Tomorrow.io (formerly ClimaCell)	A venture-backed weather company that started by using wireless signals as sensors.
2020–22	KIT with Ericsson (Germany); SMHI with Ericsson (Sweden, OpenMRG)	Thousands of links processed in near real time; open datasets and open-source tools (pycomlink).

JioMausam: one platform, three ways to earn

JioMausam is software that turns network logs into a live weather map, then turns the map into decisions: payouts, alerts and forecasts.



Why this design wins



Mausam Kavach: insurance that pays itself

Parametric insurance pays when a measured event happens, not after someone proves a loss. That removes surveyors, paperwork and fraud. It is the only way a ₹300 payout can be profitable to process.

STEP 1 - BUY

At recharge

"Add Mausam Kavach for ₹29/month" in MyJio or at a retailer. One tap, no forms. KYC is already done for the SIM.

STEP 2 - COVERED

Location-aware

Cover follows where the person works, using network location with their consent, aggregated to the ~1 km cell.

STEP 3 - TRIGGER

The network measures

e.g. ≥10 mm of rain within 3 hours, or a heat index above the local extreme, in that person's cell.

STEP 4 - PAYOUT

JioPay, same day

₹300 credited automatically, with an SMS or WhatsApp message in their language explaining the reading.

Product sheet (illustrative, to be priced with an insurer)

ITEM	DESIGN
Price	₹29 per month as a recharge add-on. Seasonal variants (monsoon, summer) for farm workers.
Rain trigger	≥10 mm in any 3-hour window in the policyholder's ~1 km cell (the level at which outdoor work and two-wheeler deliveries largely stop).
Heat trigger	Heat index above the cell's local extreme (e.g. its 98th percentile) between 11:00 and 16:00.
Payout	₹300 per trigger day (about a third to half of a day's earnings), capped at a set number of days per year.
Underwriting	Licensed insurer or reinsurer through Jio Financial Services' insurance partners. Jio is distributor and data provider.
Regulatory path	IRDAI Regulatory Sandbox for the pilot, then a filed parametric product.

How the price is set

Premium = expected payouts ÷ target loss ratio.

If a typical cell has about 0.7 trigger days a year, expected payout is $0.7 \times ₹300 = ₹210$. At a 60% loss ratio the premium is ₹350 a year, about ₹29 a month.

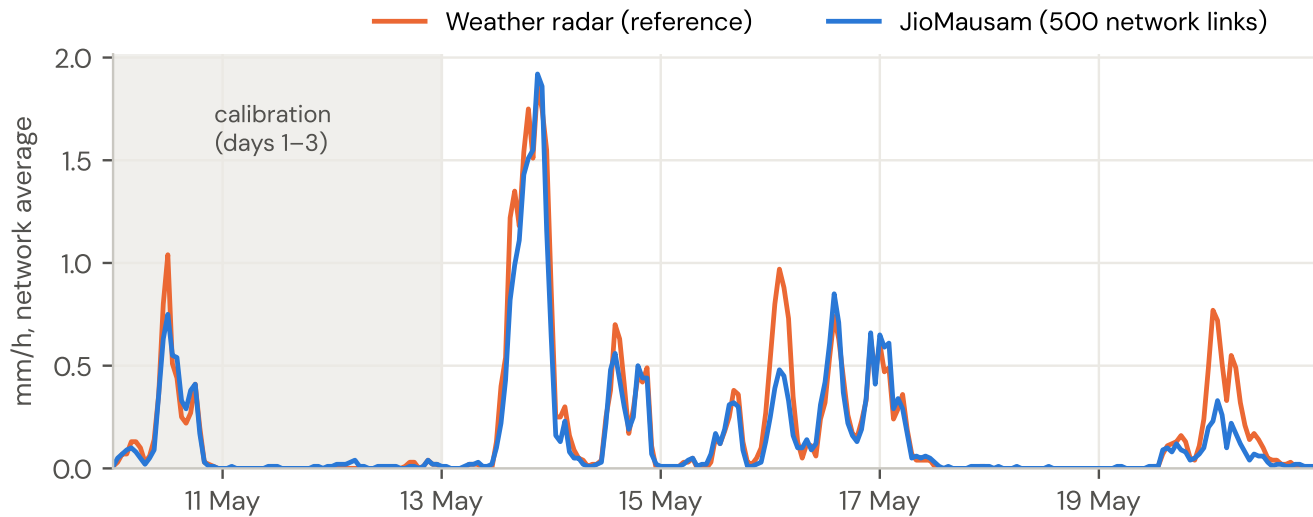
Triggers are set per cell from historical data, so a Mumbai rider and a Jaipur vendor pay the same but are protected against their own local extremes.

Why customers will trust it

- The trigger is published before you buy.
- Payout arrives without asking.
- The app shows the reading that triggered it: "12.4 mm between 2 and 5 pm near you."
- If a gig platform buys it for riders, it costs the rider nothing.

We built it and tested it on real network data

We ran the full rain pipeline on **500 commercial microwave links** from a real mobile network: 1-minute signal levels, 10–20 May 2018, 6.5–38.9 GHz, 0.5–28.6 km long. The data is the public OpenSense/pycomlink sample from Germany, with coordinates anonymised by the provider. We scored every result against the German Weather Service’s radar. We calibrated **one** scale factor ($\times 1.17$) on days 1–3 and tested on days 4–11, which the method had never seen.



Network-average rain, hour by hour. Blue is JioMausam from network links, orange is weather radar. Correlation on the held-out days: $r = 0.95$.

Results on held-out days

CHECK AGAINST RADAR	RESULT
Network-average rain, hourly correlation	0.95
Single link, median hourly correlation	0.88
Links correlating above 0.7	91%
Rainy days detected (≥ 0.1 mm) · false alarms	91% · 4.6%
Wet hours detected (≥ 1 mm/h) · false alarms	61% · 15.7%
Rain map (4 km cells) vs radar map	$r = 0.78$
Rain totals vs radar	18% lower

Payout backtest (≥ 10 mm in 3 hours)

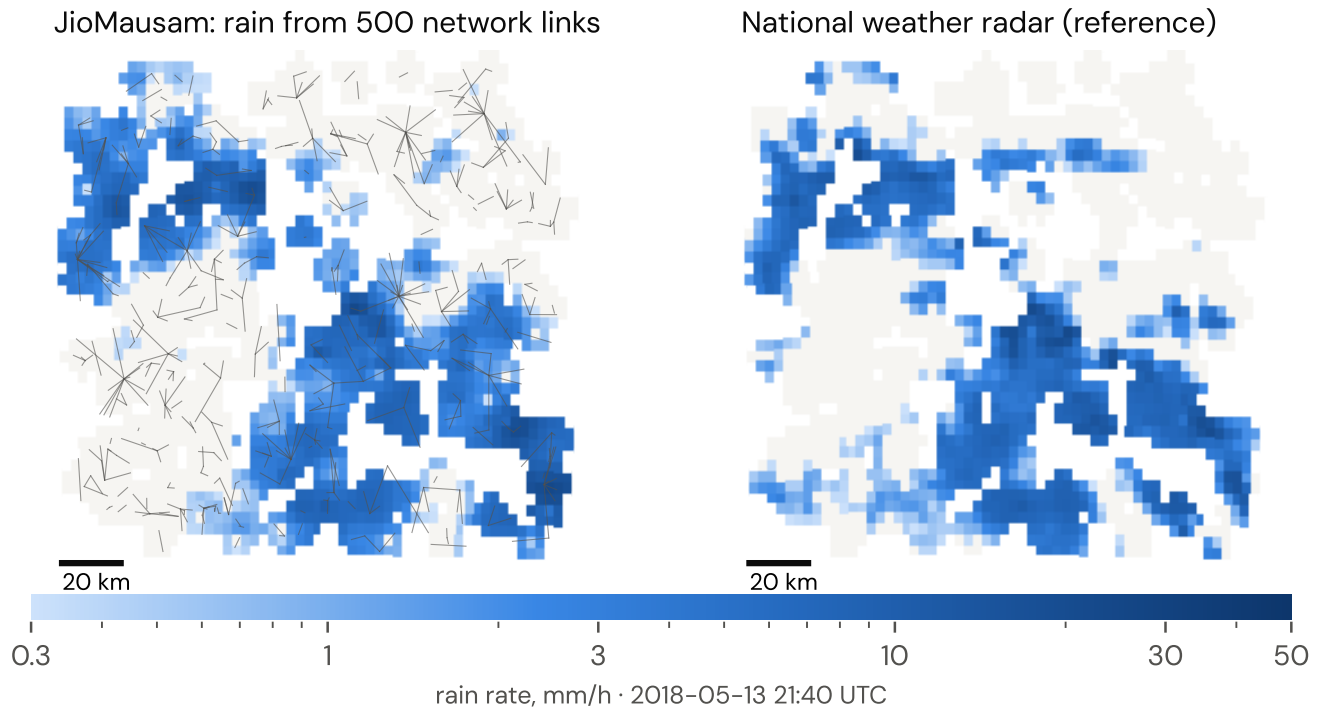
For every ~4 km cell and every held-out day (14,256 cell-days), we asked: would Kavach have paid, and would radar have paid?

Same pay / no-pay decision as radar	94.6%
Radar trigger events JioMausam also caught	68% of 1,639
JioMausam triggers radar did not confirm	18%

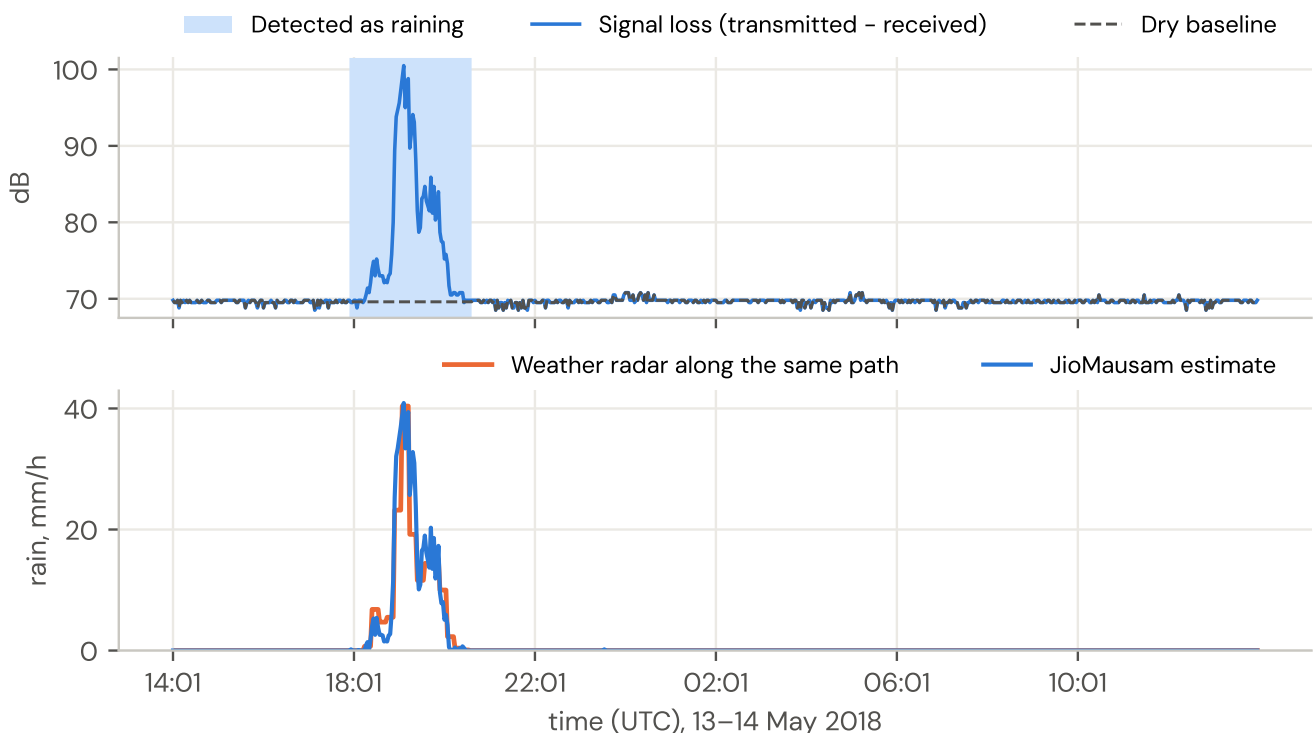
Honest reading: agreement is high partly because most cell-days are dry. The number to improve is events caught. The pilot fixes this by fusing in more links, IMD gauges and radar (see page 14).

Inside the JioMausam Control Room

The prototype is an interactive web app with four views, all running on the real data: a live map that replays the 13–14 May storm, a one-link view of the science, an accuracy dashboard, and the Kavach trigger engine paying 1,000 simulated policyholders.



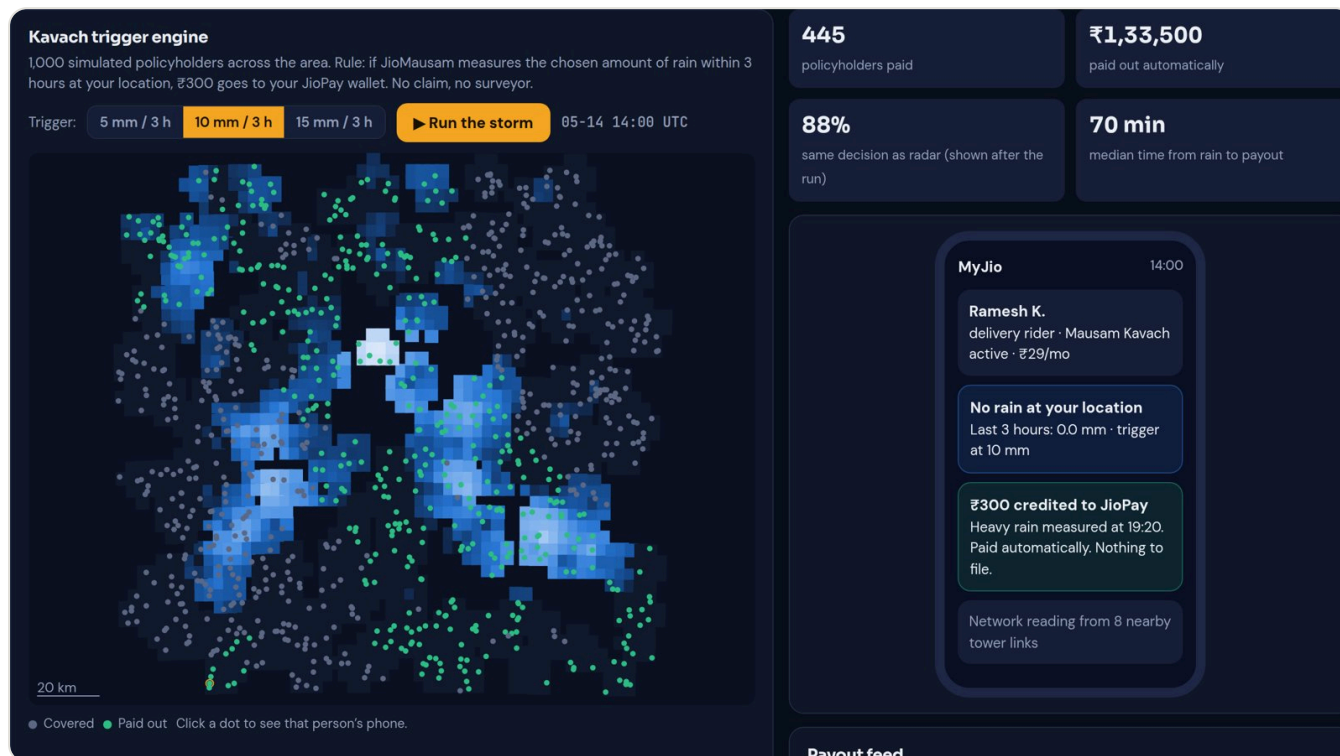
The same moment of the storm seen two ways. Left: rain estimated only from network links (grey lines). Right: the national weather radar. JioMausam captures the storm's shape and intensity with no radar at all.



One link, 7.4 km at 23 GHz. Top: when rain arrives, signal loss jumps above the dry baseline and the pipeline flags the period as wet. Bottom: the rain rate it computes tracks the radar along the same path ($r = 0.97$ during the storm).

The Kavach engine in action

The Kavach view replays the storm for 1,000 simulated policyholders. Whenever JioMausam measures the trigger amount of rain within 3 hours at someone's location, ₹300 is sent to their JioPay wallet and the payout appears in the feed. Click any dot to see that person's phone.



In this storm, with the 10 mm in 3 hours rule, 445 of 1,000 policyholders were paid ₹1,33,500 in total, a median of 70 minutes after their rain began. Switching the rule to 5 or 15 mm shows the trade-off between more payouts and fewer false triggers.

The four views, and what to say about each

VIEW	WHAT IT SHOWS	ONE LINE FOR THE JURY
Live rain map	The storm replayed every 10 minutes: network links on the left, weather radar on the right.	"Left is our network, right is a radar. Same storm, no radar needed."
How one link sees rain	Signal loss on one real link, the rain it computes, and the radar along the same path.	"This is the physics working on one link: $r = 0.97$."
Accuracy	11 days of network-average rain, every link-hour plotted against radar, and a scorecard.	"Tested on days the model never saw."
Kavach payouts	Trigger rule, payouts, phone screen and payout feed.	"About an hour after the rain starts, ₹300 is in the wallet. Nothing to file."

Demo tip: open the prototype in a browser tab before the call and pre-select the Live rain map. In the presentation, share that tab, press "Replay the storm", then switch to Kavach payouts and press "Run the storm". It takes under 40 seconds.

Business model and unit economics

The sensor costs nothing extra, and so does distribution, because people already recharge. Revenue comes from premiums, data and savings inside Reliance.

Kavach at different adoption levels (illustrative)

SHARE OF JIO'S 500M USERS	POLICIES	ANNUAL PREMIUM (₹29 × 12)	PAYOUTS AT 60% LOSS RATIO	LEFT FOR INSURER, JIO AND OPS
0.5%	2.5M	₹87 Cr	₹52 Cr	₹35 Cr
1%	5M	₹174 Cr	₹104 Cr	₹70 Cr
3%	15M	₹522 Cr	₹313 Cr	₹209 Cr

Jio's share comes from a distribution fee and a data licence fee to the underwriter. It does not carry the insurance risk.

REVENUE 1

Premiums (B2C and B2B2C)

Individuals at recharge, and gig platforms buying for their riders. One platform deal can cover lakhs of workers at once.

REVENUE 2

Mausam API (B2B and B2G)

Subscriptions for insurers pricing parametric products, state disaster authorities, municipal flood cells, agri-input and logistics firms.

REVENUE 3

Customer zero (internal)

Less waste and better stocking in JioMart, fewer solar forecast penalties for New Energy, fewer outages from rain fade for Jio.

Cost structure

- **Sensing:** close to zero, because the logs already exist. Cost is data engineering and storage.
- **Platform:** a small data science and ML team plus cloud compute (can run on Reliance's own data centres).
- **Payouts:** carried by the insurer and reinsured. Parametric design removes claims handling.
- **Customer acquisition:** near zero. It is offered inside an existing transaction (the recharge) that happens every month.

Why the economics work when micro-insurance usually fails: traditional micro-insurance dies on distribution and claims costs. Here distribution rides on the recharge and claims are automatic, so most of the premium can go back to customers.

Why only Reliance can do this

LAYER	WHAT IT NEEDS	RELIANCE ASSET
Sense	A national network with links and sites everywhere people live	Jio network, lakhs of sites, every district
Reach	A monthly touchpoint with hundreds of millions of people	500M+ subscribers, monthly recharge, MyJio
Pay	Instant, low-cost money transfer	JioPay, JioFinance (UPI rails)
Insure	A licensed underwriter and reinsurance	Jio Financial Services and its insurance partners
Use	Businesses whose costs depend on the weather	JioMart, Reliance Retail, Reliance New Energy, Jio itself

Who else is in the space, and what they lack

PLAYER	WHAT THEY DO	WHAT THEY LACK
IMD (public)	National radar, stations, forecasts	Street-level density; no payout product
Private weather firms	Forecasts and data products, often from public and satellite data	Their own dense sensor network in India; distribution
Tomorrow.io	Global weather intelligence, started with wireless-signal sensing	Access to Indian operator data; Indian distribution
Insurers and insurtechs	Parametric crop and heat pilots	Trigger data at street scale; cheap distribution
Other telecoms	Similar networks	Payments, insurance, retail and energy under one roof; scale of Jio's base

The flywheel: more subscribers mean more sites and phones, which mean better maps. Better maps mean fairer triggers, which bring more Kavach customers, who bring more data and more reasons to stay on Jio.

Bigger picture

Mobile networks that also sense their environment are a direction the 6G research community calls “integrated sensing and communication”. JioMausam is a practical first product on that path, using what 4G and 5G networks already record today.

Pilot and roadmap

WHEN	MILESTONE	WHAT HAPPENS
Days 0–90	Two-city pilot	Rain in Vidarbha (monsoon farm belt), heat in Ahmedabad (heat action plan city). Pull link and site logs, run the pipeline, and validate against IMD radar, weather stations and our own low-cost gauges at a few sites.
Months 4–9	IRDAI sandbox	Kavach with an insurance partner for 50,000 people via one gig platform and MyJio opt-in. Measure payout accuracy and customer trust.
Months 9–18	Ten cities and API	Mausam API for insurers and state disaster authorities. JioMart and New Energy go live as internal users.
Year 2+	National map	All districts. Open a public tier for disaster response. License the method to telecoms in other monsoon countries.

Pilot success criteria

≥ 0.8

hourly correlation with IMD station rain in pilot cells (prototype: 0.88 median per link)

$\geq 80\%$

of heavy-rain trigger events caught, with fewer than 15% false triggers

$< 6\text{ h}$

from trigger to money in the wallet, for 95% of payouts

What we need from Reliance

- Access to link performance logs (received and transmitted power) for the pilot regions, anonymised and aggregated by Jio's own team.
- A product owner in Jio Financial Services and an insurance partner for the sandbox.
- A MyJio placement for the opt-in, and an introduction to one gig platform.
- A small core team: 2 data scientists, 1 radio-network engineer, 1 product manager, 1 actuary (from the partner).

Partners to bring in

IMD and Mission Mausam (validation and public good), state disaster management authorities (alerts), municipal corporations (urban flooding), gig platforms (B2B2C cover), and research groups that already run this method (KIT, SMHI) for a fast start.

What could go wrong, and how we handle it

RISK	WHY IT MATTERS	HOW WE HANDLE IT
Jio backhaul is mostly fibre Medium	Fibre has no rain signal, so fewer microwave links means less rain coverage.	Use the microwave links that remain (rural, hilly and E-band sites). Add 4G/5G radio quality and phone data. Fuse with IMD radar and gauges. Map coverage honestly with a confidence score per cell.
Basis risk Medium	A worker loses income but the trigger doesn't fire, or the reverse.	Small cells, multi-source fusion, triggers tuned per cell, a published rule, and a goodwill review queue for disputes. The prototype already agrees with radar 94.6% of the time before any fusion.
Underestimation Low	Prototype totals run 18% below radar.	Known effect of the simple baseline. Continuous calibration against gauges, and better wet-antenna models from the literature.
Privacy Medium	Location and phone data are personal data under the DPDP Act, 2023.	Rain uses network equipment data, not personal data. Heat uses opt-in, on-device aggregation, reported only per area. Clear consent at purchase. Data stays inside Reliance.
Regulation Low	Insurance products need IRDAI approval.	Start in the IRDAI Regulatory Sandbox with a licensed partner. Jio acts as distributor and data provider only.
Adoption and trust Medium	Low insurance awareness among the target users.	Sell through platforms that pay for riders. The first payout is the best marketing. Explain in local languages over WhatsApp and IVR.
Gaming Low	People buying cover only when a storm is forecast.	Waiting period of 7 days after purchase. Monthly auto-renewal with the recharge.
Climate trend Medium	Extremes are becoming more frequent, which raises payouts.	Reprice yearly per cell. Reinsurance. The data itself becomes more valuable as extremes grow.

Our 10 minutes with the jury

Ten minutes: the 3-minute video plays first, then 3 minutes of our presentation, then 4 minutes of Q&A. Everyone joins from a separate room with the camera on, named **TeamName_YourName**. Upload the deck via Present → Slides → Your computer or Google Slides.

0:00–3:00 · The video

The new 3-minute video (2:54) covers hook, problem, insight, **proof from our prototype**, product, business, why Reliance, and roadmap. While it plays we join the call, turn cameras on and get the deck ready to share.

3:00–6:00 · The presentation (don't repeat the video, go deeper)

TIME	SLIDE	WHAT TO SAY
0:00	Live demo of the Control Room	"This is real data from 500 network links. Watch the storm. Left is our network, right is the radar." Press Replay.
0:40	Results vs radar	Three numbers: $r = 0.95$; 91% of rainy days caught; 94.6% payout agreement. Mention the held-out days.
1:20	Kavach journey	Buy at recharge, trigger, ₹300 in JioPay. Show the phone screen in the Kavach tab.
1:50	Unit economics	₹174 Cr at 1% adoption, near-zero acquisition cost, three revenue layers.
2:20	Pilot and ask	90-day pilot, success criteria, what we need from Reliance.
2:50	Close	"We don't build sensors. We become the sensor."

6:00–10:00 · Q&A

Split the questions by owner

- **Science and prototype:** how it works, accuracy, data.
- **Product and customer:** Kavach, trust, adoption.
- **Business and strategy:** economics, moat, roadmap.
- **Risk and regulation:** privacy, IRDAI, basis risk.

How to answer

- Answer in one sentence first, then one piece of evidence.
- Keep each answer under 30 seconds. Four minutes means 5–7 questions.
- If you don't know, say what the pilot will measure to find out.
- Never argue with a judge. "Great point, here is how we'd handle it."

Jury Q&A bank

1. How accurate is it, really?

On 500 real links and days the model never saw, network-average rain matched radar at $r = 0.95$, the median single link at $r = 0.88$, and it caught 91% of rainy days with 4.6% false alarms. Peer-reviewed studies in the Netherlands and Germany show similar results at national scale.

2. Isn't Jio's backhaul mostly fibre?

Much of it, yes, and fibre doesn't sense rain. But microwave links remain at many rural, hilly and fast-rollout sites. We add 4G/5G radio quality data and phone sensors, then fuse with IMD radar and gauges. The pilot's first job is to measure real coverage.

3. Why not just use IMD or satellite data?

We do use them, as inputs to the fusion. But radar is sparse, satellites see clouds rather than rain at the street, and gauges are points. The network adds density exactly where people live, every minute, at no extra cost.

4. Your data is from Germany. Does it work in India?

The physics is the same everywhere. Tropical rain is heavier, which makes the signal stronger and easier to detect. The pipeline is open-source and has also been used in Africa. The pilot validates it on Indian links against IMD.

5. What about privacy?

Rain needs no personal data: it uses network equipment logs. Heat uses opt-in phone readings aggregated by area. Payouts use the policyholder's own consented location. This is designed to the DPDP Act, 2023.

6. What is basis risk and how do you reduce it?

It is the gap between the trigger and the real loss. We cut it with ~1 km cells, multi-source fusion, triggers set per location, and a review queue. Our backtest already shows 94.6% agreement with radar decisions.

7. Why would a rider pay ₹29 a month?

It is the price of a cup of tea, it comes with the recharge, and it pays without asking. Better still, platforms buy it for riders, as some states' gig-worker welfare laws push platforms to fund protection.

8. Who takes the insurance risk?

A licensed insurer and reinsurer through Jio Financial Services' partners. Jio earns distribution and data fees and carries no underwriting risk.

9. How do the numbers add up?

1% of 500M users is 5M policies $\times$ ₹348 a year = ₹174 Cr premium. At a 60% loss ratio, ₹104 Cr goes back as payouts. Acquisition cost is near zero because it rides on the recharge.

10. Can't Airtel or Vi copy this?

They can sense rain, but they don't have a payments app, an insurance arm, a retail chain and an energy business that all monetise the data. Speed matters too: the first network to map India owns the reference dataset.

11. How long until it's live?

Ninety days to a two-city map, nine months to a sandbox product, and about 18 months to ten cities with a paid API.

12. What does Reliance need to invest?

A small team (about five people), access to existing logs, cloud compute and a MyJio placement. No hardware capex. That is the point.

Harder questions

13. Your totals are lower than radar. Isn't that a problem?

Yes, the simple baseline underestimates by about 18%. It is well understood and fixed by continuous calibration against gauges. Triggers are also set from our own historical readings, so a consistent bias does not change who gets paid.

14. What if the network goes down in a storm?

A link outage is itself a signal: total signal loss in heavy rain shows up as an extreme reading, and nearby links fill the gap. The map shows lower confidence for that cell, and payouts can fall back to radar.

15. How do you stop people gaming the cover?

A 7-day waiting period, monthly auto-renewal and location consistency checks. Payouts are capped. Nobody can make it rain on a tower.

16. Why heat, if your prototype is only rain?

Heat is the bigger income killer for outdoor workers in North and West India. The science is published. It is our module two, built on the same platform. We led with rain because we could prove it on real data now.

17. What's the social impact?

Income protection for workers who have none, earlier flood and heat warnings for cities, and better data for India's climate adaptation, from infrastructure the country has already paid for.

18. What's the biggest risk?

Coverage, if too few microwave links remain in a city. That is why the first 90 days measure exactly that, and why the design fuses several sources rather than depending on one.

19. Why should this be a Reliance business and not a CSR project?

Because it earns: premiums, data licences, and savings for JioMart, New Energy and Jio. And it deepens loyalty, since Kavach customers are less likely to switch networks.

20. If you had ₹10 crore, where would it go?

The pilot team and data platform first, then the insurance sandbox and customer education in two cities, then the API. Most of it is people, not hardware.

If you get a question you can't answer: "We haven't measured that yet. It's exactly what the 90-day pilot is designed to answer, and here's the metric we'd track."

Glossary

Microwave link (CML)	A point-to-point radio connection between two towers, usually 6–40 GHz, used to carry traffic where there is no fibre.
Attenuation / signal loss	How much weaker the signal arrives than it was sent, in decibels (dB). Rain adds extra loss.
Baseline	The link's normal dry-weather loss. Rain is measured as loss above the baseline.
Wet antenna effect	A film of water on the antenna adds loss even in light rain. We subtract it so we don't overcount.
k–R relation	The ITU formula linking loss per km to rain rate: $\text{loss} = a \times R^b$.
Correlation (r)	How closely two series move together, from 0 (unrelated) to 1 (identical pattern).
Detection rate / false alarms	Of the events radar saw, the share we also caught. Of the events we flagged, the share radar didn't see.
Parametric insurance	Insurance that pays when a measured index crosses a trigger, with no loss assessment.
Basis risk	The chance that the index and the real loss disagree.
Loss ratio	Payouts as a share of premiums.
IRDAI Regulatory Sandbox	The insurance regulator's framework for testing new products with limited customers before full approval.
DPDP Act, 2023	India's Digital Personal Data Protection Act, which governs consent and use of personal data.
PMFBY	Pradhan Mantri Fasal Bima Yojana, the government crop insurance scheme.

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- Prototype data: pycomlink example dataset (OpenSense), radar reference from DWD RADOLAN–YW.

Before the round, re-check the latest figures for IMD radar count, Jio's subscriber base and Jio Financial Services' insurance partnerships, as these change.

Voice-over script for the 3-minute video

145 words per minute, total 2:54. Each timestamp is when the slide changes. Read at a calm, confident pace and pause briefly at each slide change.

[0:00] SLIDE 1 • Hook

India's biggest weather network already exists.
It has lakhs of stations, it covers every district, and it's switched on right now.
Nobody noticed — because we call it Jio.

[0:12] SLIDE 2 • Problem

Weather in India is hyperlocal, but our data isn't.
About forty Doppler radars watch 3.3 million square kilometres.
When a cloudburst hits one ward, the people who lose the most —
farmers, delivery riders, construction workers, street vendors — lose it alone.
Heat alone could cost India thirty-four million full-time jobs by 2030.
And when insurance does exist, claims take months.

[0:37] SLIDE 3 • Insight

Here's the insight.
Rain weakens the radio links between towers —
and how much it weakens tells you how hard it's raining, minute by minute.
Smartphone batteries, averaged across a city, track air temperature.
So every tower is a rain gauge. Every phone is a thermometer.
We don't build sensors. We become the sensor.

[1:00] SLIDE 4 • Proof

Does it actually work? We built it.
We ran our prototype on 500 real microwave links for eleven days —
and checked it against national weather radar.
Network-wide, our rain readings match radar with a correlation of 0.95.
Our payout rule made the same decision as radar ninety-five percent of the time.

[1:21] SLIDE 5 • Product

Meet JioMausam —
a live, street-level weather map built from network data.
On top of it: Mausam Kavach —
parametric micro-insurance you add to your recharge for twenty-nine rupees a month.
When the network measures 45 degrees in your area, or rain heavy enough to stop work,
the payout lands in your JioPay wallet within hours.
No claim forms. No surveyor. No waiting.

[1:47] SLIDE 6 • Business

The business has three layers.
One: Kavach premiums — one percent of Jio's 500 million users is five million policies —
around 170 crore rupees a year.
Two: B2B — gig platforms buy cover for riders; insurers and states buy hyperlocal data.
Three: Reliance is customer zero —
JioMart forecasts rainy-day demand, New Energy forecasts solar output, and Jio predicts its own rain-fade.

[2:12] SLIDE 7 • Why Reliance

Why only Reliance?
A startup can't own the towers.
An insurer can't reach 500 million people.
No other telecom has a payments app, an insurance arm, a retail chain
and a giga-scale energy business under one roof.
Reliance has every layer.

[2:30] SLIDE 8 • Roadmap & close

We start with a ninety-day pilot in Vidarbha and Ahmedabad, validated against IMD gauges.
Then an IRDAI sandbox product. Then a national map.
India won't adapt to climate change only by building new infrastructure.
It'll adapt by listening to the infrastructure it already has.
This is JioMausam. Different by design.